SPACEX

Falcon 9 First Stage Landing Prediction

IBM Data Science Professional Certificate — Capstone Project

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OUTLINE

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EXECUTIVE SUMMARY

90	66.7%	83.3%	3
Falcon 9 Launches	Overall Success Rate	Best Model Accuracy	Launch Sites

Project Goal: Predict whether the Falcon 9 first stage will land successfully, enabling cost estimation ($62M vs $165M+ for competitors).

Methodology:
Collected data via SpaceX REST API → EDA with visualization & SQL → Interactive Folium maps & Plotly Dash dashboards → Trained 4 classification models with GridSearchCV.

Key Finding:
Logistic Regression, SVM, and KNN all achieved 83.3% test accuracy. Launch success increases with flight number and strongly correlates with launch site and orbit type. KSC LC-39A has the highest success rate.

INTRODUCTION

Project Background

SpaceX has revolutionized space travel by reusing the Falcon 9 first stage, reducing launch costs from $165M to just $62M.

Predicting landing success is crucial for accurate launch cost estimation — a key factor for companies bidding against SpaceX.

The first stage accounts for most of the launch cost. If it lands successfully, the rocket can be refurbished and reused, dramatically lowering the price.

Problem Statement

Can we predict if the Falcon 9 first stage will land successfully?

Key Questions:

→ Which factors influence landing success?
→ Does payload mass affect outcome?
→ Which launch site has highest success?
→ Which orbits are most successful?
→ Has success rate improved over time?
→ Which ML model best predicts landing?

METHODOLOGY

Data Collection → Wrangling → EDA → Interactive Analytics → Predictive Analysis

METHODOLOGY OVERVIEW

1

Data Collection

SpaceX REST API
& Web Scraping

→

2

Data Wrangling

Missing values,
Label creation

→

3

EDA

Visualization
& SQL Analysis

→

4

Interactive Analytics

Folium Maps
& Plotly Dash

→

5

Predictive Analysis

4 ML Models
GridSearchCV

Tools & Technologies

Python • Pandas • NumPy • Matplotlib • Seaborn • Scikit-learn • Folium • Plotly Dash • SQL (SQLite) • SpaceX REST API • BeautifulSoup • Jupyter Notebook

DATA COLLECTION — SpaceX API

1 GET Request

api.spacexdata.com
/v4/launches/past

→

2 Parse JSON

json_normalize()
to DataFrame

→

3 Enrich Data

API calls for rocket,
launchpad, payload, cores

Helper Functions for Data Enrichment

`getBoosterVersion()`	Rocket name from /v4/rockets/{id}
`getLaunchSite()`	Site name, lat, long from /v4/launchpads/{id}
`getPayloadData()`	Mass & orbit from /v4/payloads/{id}
`getCoreData()`	Landing outcome, type, flights, gridfins, legs, block, serial

Final dataset: 90 Falcon 9 launches with 17 features (Falcon 1 filtered out)

DATA WRANGLING

Handling Missing Values

PayloadMass: 5 missing values replaced with column mean

LandingPad:
26 None values retained (no landing pad used)

All other columns:
No missing values detected

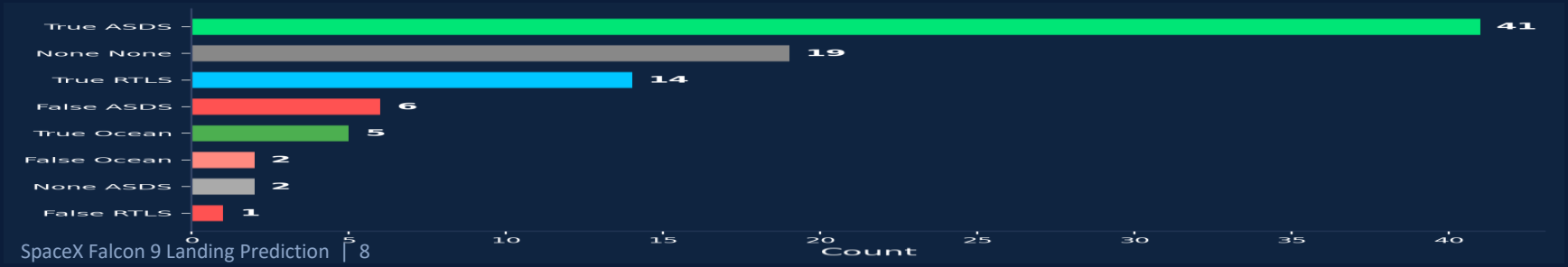
Binary Classification Label

Created 'Class' column:

Class = 1 (Success):
True ASDS, True RTLS, True Ocean

Class = 0 (Failure):
False ASDS/RTLS/Ocean, None None/ASDS

Overall success rate:
66.7%



EDA WITH VISUALIZATION

Flight Number vs Launch Site

Higher flight numbers show more success, especially at KSC LC-39A

Payload vs Launch Site

VAFB SLC-4E has no heavy payload launches (>10,000 kg)

Success Rate vs Orbit Type

ES-L1, GEO, HEO achieve 100% success; SO has 0%

Flight Number vs Orbit Type

LEO success correlates with flight count; GTO shows no pattern

Payload vs Orbit Type

Heavy payloads succeed more in Polar, LEO, and ISS orbits

Launch Success Yearly Trend

Success rate steadily increased from 0% (2010) to 87% (2020)

Feature Engineering: OneHotEncoding on categorical columns → 80 features, all cast to float64

EDA WITH SQL

10 SQL queries on SpaceX launch data using SQLite:

1. SELECT DISTINCT Launch_Site — Identified 3 unique launch sites
2. SELECT * WHERE Launch_Site LIKE 'CCA%' LIMIT 5 — Cape Canaveral records
3. SUM(Payload_Mass) WHERE Customer='NASA (CRS)' — Total NASA payload
4. AVG(Payload_Mass) WHERE Booster='F9 v1.1' — Avg F9 v1.1 payload
5. MIN(Date) WHERE Landing='Success (ground pad)' — First ground landing
6. Boosters with drone ship success & payload 4000–6000 kg
7. COUNT by Mission_Outcome — Success vs failure totals
8. Boosters carrying MAX payload (subquery)
9. Failed drone ship landings in 2015 with booster & site info
10. Ranked landing outcomes between 2010-06-04 and 2017-03-20

INTERACTIVE VISUAL ANALYTICS

Folium Interactive Maps

Task 1: Marked all launch sites with Circle markers and labeled popups

Task 2:
Color-coded outcomes — green (success) / red (failure) — using MarkerCluster

Task 3:
Calculated distances from CCAFS SLC-40 to coastline (0.86 km), highway, railway, and city using Haversine formula

Key Insight:
All sites near coast and close to equator (except VAFB in California)

Plotly Dash Dashboard

Component 1: Dropdown for launch sites — pie chart of success count

Component 2:
Range slider for payload (0–10,000 kg) — filters scatter plot dynamically

Component 3:
Scatter plot colored by Booster Version showing payload vs. outcome

Key Insight:
KSC LC-39A has highest success ratio. Payload 2000–4000 kg is the sweet spot

PREDICTIVE ANALYSIS — METHODOLOGY

Standardize

StandardScaler on
80 one-hot features

→

Train/Test Split

80/20 split
(72 train / 18 test)

→

GridSearchCV

10-fold CV for
hyperparameter tuning

→

Evaluate

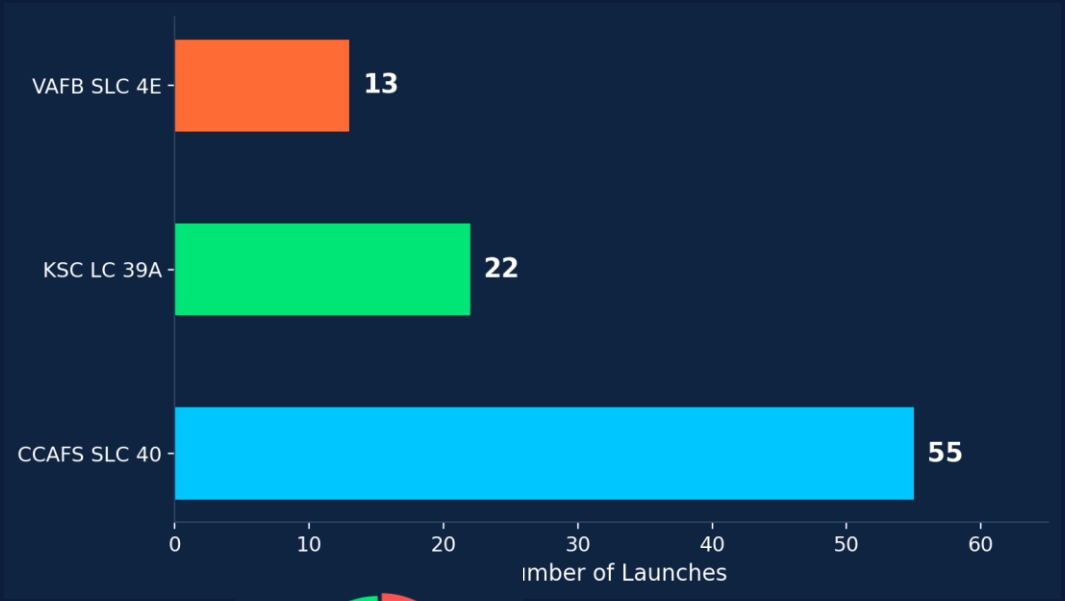
Test accuracy &
confusion matrix

Model	Hyperparameters Tuned	Best Parameters	CV Acc
Logistic Regression	C: [0.01, 0.1, 1]	C=0.01, penalty=l2	84.6%
SVM	C, gamma, kernel	C=1.0, gamma=0.032, sigmoid	84.8%
Decision Tree	criterion, max_depth, splitter	entropy, depth=12, random	87.5%
KNN	n_neighbors, algorithm, p	n=10, auto, p=1	84.8%

RESULTS

EDA Visualizations • SQL Analysis • Interactive Maps • Dashboard • Prediction

RESULTS — Launch Site Analysis

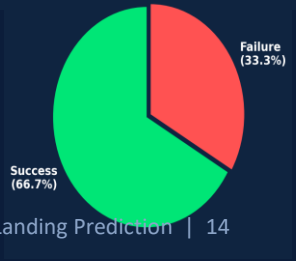


Key Findings

CCAFS SLC 40 dominates with 55 launches (61%), mainly early missions

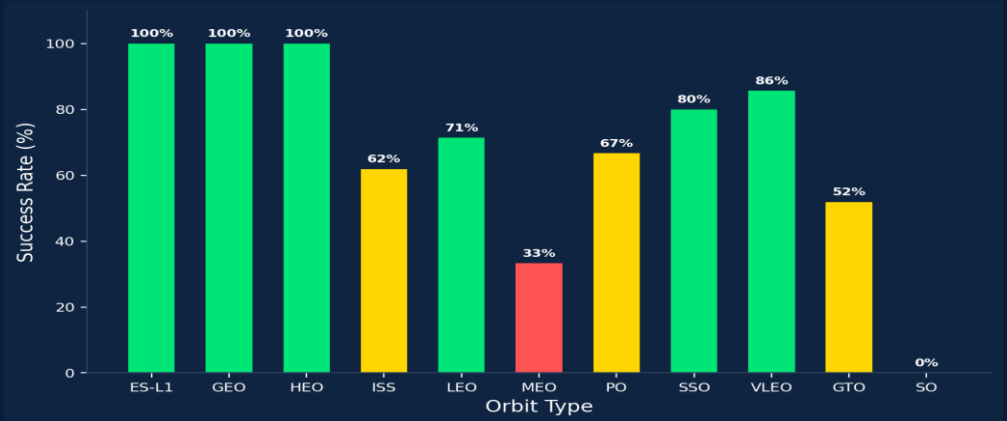
KSC LC-39A
has highest success rate, with later, more reliable missions

VAFB SLC-4E
is for polar orbits only, 13 flights, no heavy payloads >10,000 kg



Overall: 66.7% success rate across 90 Falcon 9 launches

RESULTS — Orbit Success & Yearly Trend



Orbit Success: ES-L1, GEO, HEO achieve 100%. GTO (27 launches) has ~52%. SO is 0%.

Yearly Trend: Success improved from 0% (2010) to ~87% (2020). Steepest growth in 2015–2017, coinciding with reusability tech advances.

RESULTS — SQL Analysis Highlights

Unique Launch Sites

CCAFS SLC-40, KSC LC-39A,
VAFB SLC-4E

Total NASA (CRS) Payload

45,596 kg
across all CRS missions

Avg F9 v1.1 Payload

2,928.4 kg
average per launch

First Ground Landing

2015-12-22
(historic milestone!)

Max Payload Boosters

F9 B5 B1048.4,
F9 B5 B1049.4, and more

Ranked Outcomes (2010–2017)

No Attempt: 10, Failure: 5,
Success (drone): 5, Ground: 3

RESULTS — Interactive Map with Folium

Launch Site Markers

4 sites pinned on global map. Florida east coast cluster (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and California (VAFB SLC-4E).

All sites near coastline for abort safety

Success/Failure Map

MarkerCluster with green/red markers. KSC LC-39A shows highest density of green. CCAFS has mix of early failures and later successes.

KSC LC-39A is the most successful site

Proximity Analysis

CCAFS SLC-40 is 0.86 km from coastline. PolyLines to nearest highway, railway, city. All sites have nearby infrastructure.

Strategic coastal + infrastructure positioning

Coordinates: CCAFS LC-40 (28.56°N) | CCAFS SLC-40 (28.56°N) | KSC LC-39A (28.57°N) | VAFB SLC-4E (34.63°N)

RESULTS — Plotly Dash Dashboard

Pie Chart — All Sites

KSC LC-39A contributes the most successful launches, followed by CCAFS SLC-40.

KSC LC-39A: Highest success count

Pie Chart — Success Ratio

KSC LC-39A shows ~76.7% success rate, significantly outperforming other sites.

KSC LC-39A: ~76.7% success rate

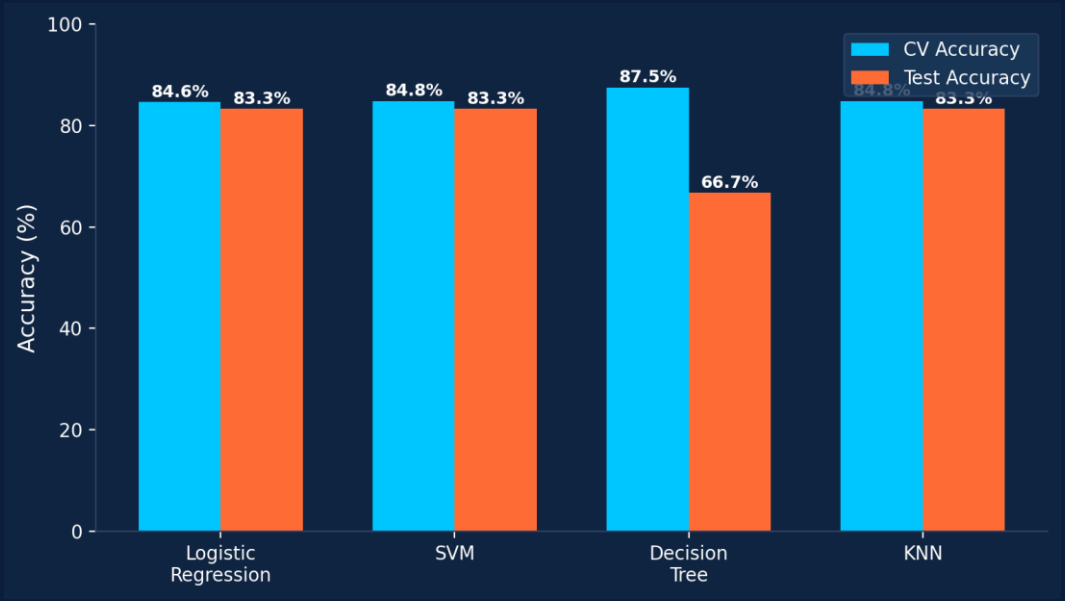
Scatter — Payload vs Outcome

Payload 2000–4000 kg has highest success. FT and B5 boosters dominate heavy successes.

Mid-range payloads most successful

Innovative Insight: Very light (<1000 kg) and very heavy (>9000 kg) payloads both show lower success than mid-range (2000–6000 kg), suggesting an optimal payload 'sweet spot' for landing, likely tied to fuel margin and re-entry dynamics.

RESULTS — Classification Accuracy



Model Comparison

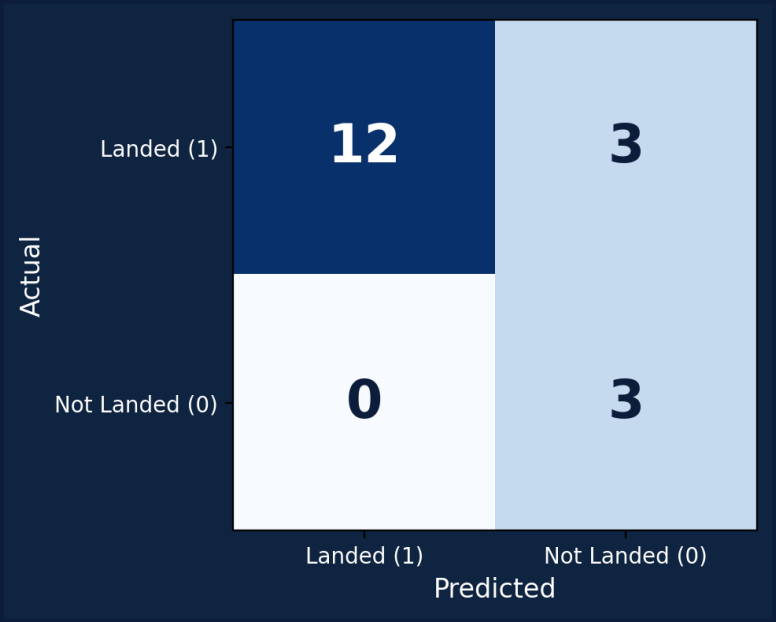
Three models tied at 83.3%:
LR, SVM, KNN

Decision Tree: highest CV (87.5%) but worst test (66.7%) —
overfitting.

Recommended: Logistic Regression
→ Competitive accuracy (83.3%)
→ High interpretability
→ Fastest inference time
→ Stable generalization

Innovative Insight: Small test set (18 samples) limits differentiation — 1 misclassification changes accuracy by ~5.5%. Bootstrap CI or larger test sets would improve robustness.

RESULTS — Confusion Matrix (Best Model)



Precision: 80% | Recall: 100% | F1-Score: 88.9% | Accuracy: 83.3%

Confusion Matrix Breakdown

12 — **True Positive**
Correctly predicted successful landing

3 — **False Positive**
Predicted success, actually failed

3 — **True Negative**
Correctly predicted failed landing

0 — **False Negative**
No missed successful landings!

CONCLUSIONS

Success Rate Trends Upward

From 0% (2010) to ~87% (2020), demonstrating continuous reusability improvement.

Launch Site Matters

KSC LC-39A achieves ~76.7% success. All sites near coastlines for safety.

Orbit Type Affects Outcomes

ES-L1, GEO, HEO: 100% success. GTO ~52%, SO 0%.

Payload Sweet Spot Exists

2,000–6,000 kg payloads show highest success rates.

ML Models Achieve 83.3%

LR, SVM, KNN tied. DT overfits. LR recommended for interpretability.

Cost Estimation is Feasible

Predicting landing success enables $62M vs $165M cost estimation for competitors.

THANK YOU

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Appendix — Notebook References

1. Data Collection (SpaceX API) – jupyter_labs_spacex_data_collection_api.ipynb
2. Data Wrangling – labs-jupyter-spacex-Data_wrangling.ipynb
3. EDA with Visualization – edadataviz.ipynb
4. EDA with SQL – jupyter_labs_eda_sql_coursera_sqllite.ipynb
5. Interactive Maps (Folium) – lab_jupyter_launch_site_location.ipynb
6. Machine Learning Prediction – SpaceX_Machine_Learning_Prediction_Part_5.ipynb